Geographic Information Retrieval (2026W)

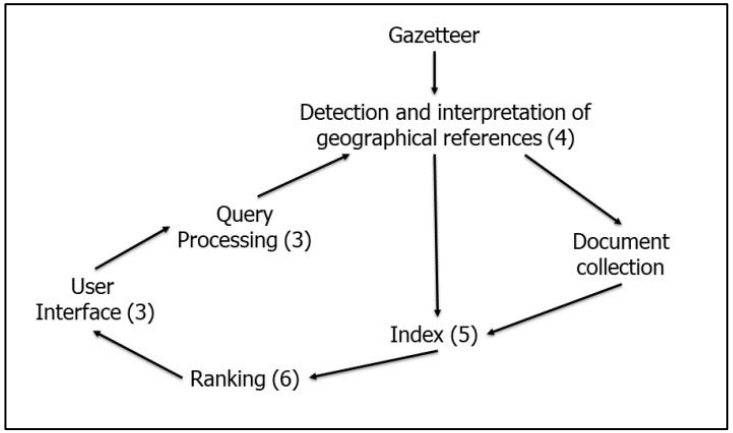
Yingjing Huang & Zilong Liu
yingjing.huang@univie.ac.at
zilong.liu@univie.ac.at

Lecture 3

Geoparsing II

Recap

Why geoparsing? To match queries with relevant documents for GIR.



Geoparsing: Detecting place names from text.

Input: French company Aeroports de Paris International is the design consultant for the work. The company also operates Charles de Gaulle airport in Paris, where part of a terminal building's roof collapsed in May, killing four people.
Output: French company <org>Aeroports de Paris International</org> is the design consultant for the work. The company also operates <loc>Charles de Gaulle airport</loc> in <loc>Paris</loc>, where part of a terminal building's roof collapsed in May, killing four people.

How?

Tokenization + one of the following approaches:

1. **PoS tagging + gazetteer lookup**
2. **Rule-based methods**
3. **Machine learning**

spaCy

“Danube” PERSON?

Vienna GPE is the capital, most populous city, and one CARDINAL of the nine CARDINAL states of Austria GPE . It is Austria GPE 's primate city, with just over two million CARDINAL inhabitants. Its larger metropolitan area has a population of nearly 2.9 million CARDINAL , representing nearly one-third CARDINAL of the country's population. Vienna GPE is the cultural, economic, and political centre of the country, the fifth ORDINAL -largest city by population in the European Union ORG , and the most populous of the cities on the river Danube PERSON .

Learning Objectives: Geoparsing Evaluation

- spaCy is a **general-purpose NER** package, not specifically designed for geoparsing.
- Even with machine learning techniques, geoparsing is **not a completely solved** problem.
- Any tool that **appears** to perform well still needs to be carefully evaluated.

Evaluation Metrics

Why Evaluation Metrics?

- **Evaluation metrics**: quantitative measures used to assess how well a machine learning model performs on a given task
- We need to grade the model performance by providing a number.

Sample Evaluation Data

- We will use the following example to learn evaluation metrics.
- **Questions**: What are place names here? What are (in)correct predictions?



Positive

- **Positive**: a target class that a machine learning model is designed to predict
- **“Target class” in geoparsing**: place name

Positive (Cont'd)

Ground-truth positives:

- ❏ "London" (1)
- ❏ "Paris" (1)
- ❏ "Lake Geneva" (1)
- ❏ "Seattle" (1)

Predicted positives:

- ❏ "Jordan" (1)
- ❏ "London" (1)
- ❏ "Paris" (1)
- ❏ "Washington" (1)
- ❏ "Seattle" (1)



Positive (Cont'd)

True positives - correct positives predicted by a model:

- "London" (1)
- "Paris" (1)
- "Seattle" (1)

False positives - incorrect positives predicted by a model:

- "Jordan" (1)
- "Washington" (1)

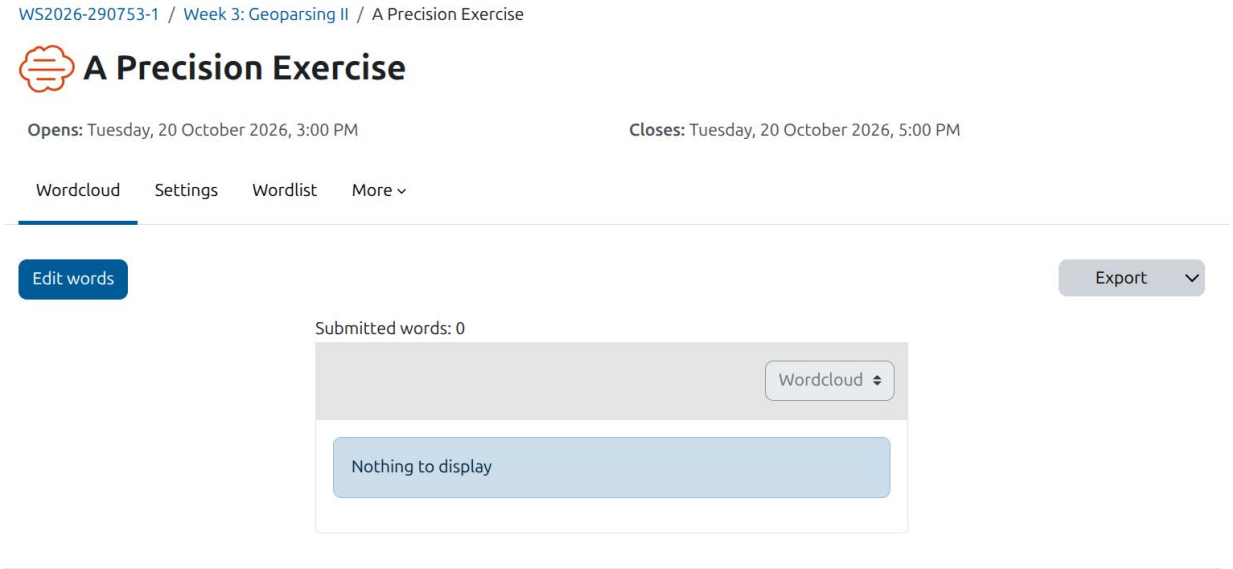
Our 1st Metric: Precision

- **TP**: True positives
- **FP**: False positives
- **In our example**:
  - TP = 3
  - FP = 2
  - Precision = 3/(3+2) = 0.6

$$\text{Precision} = \frac{TP}{TP + FP}$$

Our 1st Exercise

- Suppose a model predicts that there are 10 place names.
- Among these predictions, there are 8 correct ones.
- What is the Precision here? [Tip: Try to figure out TP and FP too.]



Negative

- **Negative**: a non-target class that a machine learning model is also designed to predict
- **“Non-target” class in geoparsing**: other nouns (e.g., person) and other parts of speech (e.g., adjectives)

Negative (Cont'd)

Ground-truth negatives:

- ...

Predicted negatives:

- ...



Negative (Cont'd)

True negatives - correct negatives predicted by a model:

- ...

False negatives - incorrect negatives predicted by a model:

- "Lake Geneva" (1)



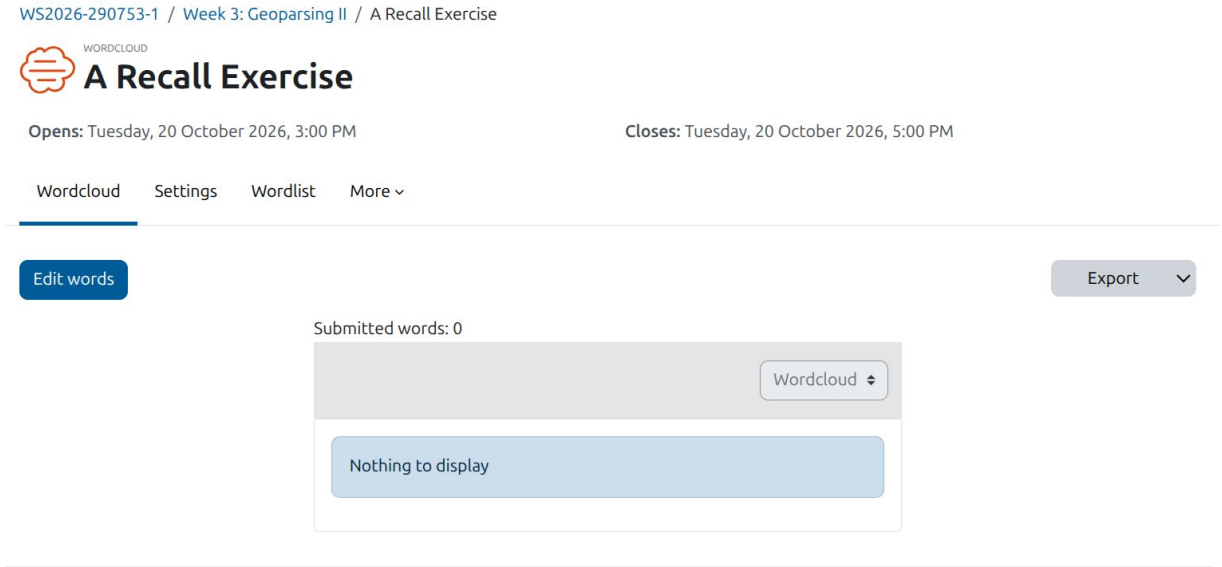
Our 2nd Metric: Recall

- **TP**: True positives
- **FN**: False negatives
- **In our example**:
  - TP = 3
  - FN = 1
  - Recall = 3/(3+1) = 0.75

$$\text{Recall} = \frac{TP}{TP + FN}$$

Our 2nd Exercise

- Suppose there are 12 ground-truth place names.
- A model misses 4.
- What is the Recall here? [Tip: Try to figure out TP and FN too.]



The Rationale Behind

- **Precision** - Of all positives predicted by a model, how many are also annotated positive?
- **Recall** - Of all positives annotated by humans, how many are also predicted positive?

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Our 3rd Metric: F1 Score

- **F1 score**: the harmonic mean of Precision and Recall
- **"Harmonic mean"**: a type of average calculated by dividing the number of values in a data set by the sum of their reciprocals

$$H(x_1, x_2, \ldots, x_n) = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}.$$

- **In our example**:
  - Precision = 0.6
  - Recall = 0.75
  - F1 = 2/(1/0.6 + 1/0.75) ≈ 0.67

$$F_1 = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$$

Beyond Evaluation Metrics

Evaluation -> Observation

- Evaluation metrics provide **macro-level** insights into geoparsing performance.
- Evaluation is also about observing and understanding model behavior.
- Micro-level evaluation: Examining individual predictions to understand where a geoparser succeeds or fails.

Open Research Questions

- **Question 1**: How does the input context influence geoparsing behavior?
- **Question 2**: How will the choice of language data and pipeline influence geoparsing behavior?
- **Question 3**: How will the model size influence geoparsing behavior?
- …

…, [*neighboring word*], [*target word*], [*neighboring word*], …

…, [*english word*], [*english word*], [*english word*], …

an English/German spaCy pipeline?

a smaller/larger spaCy pipeline?

Sample Place Name: Kärntner Straße

- We will use it as our example.



https://www.wien.info/en/now-on/christmas/vienna-christmas-lights-362996

Input Context

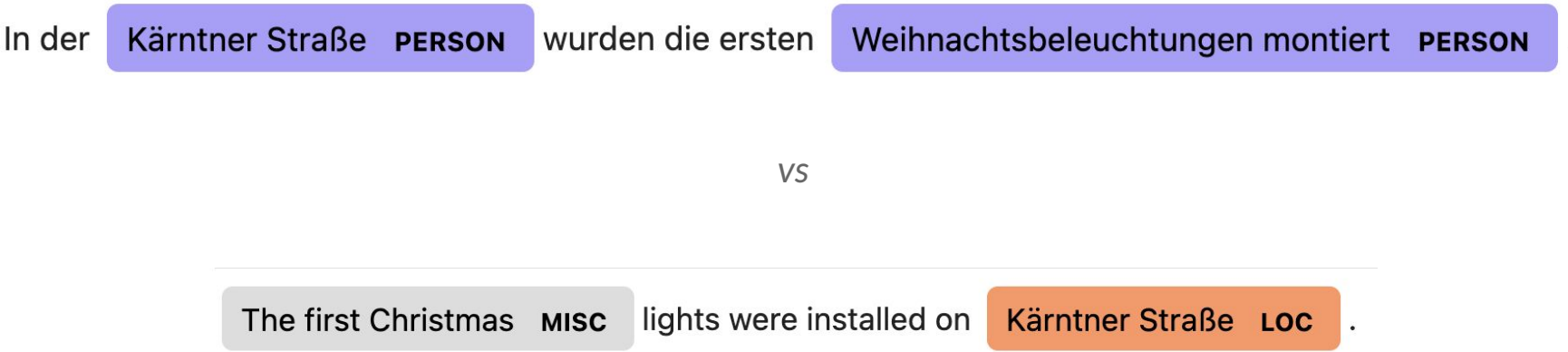
Kärntner Straße

vs



Example: How spaCy's small English pipeline works for "Kärntner Straße"

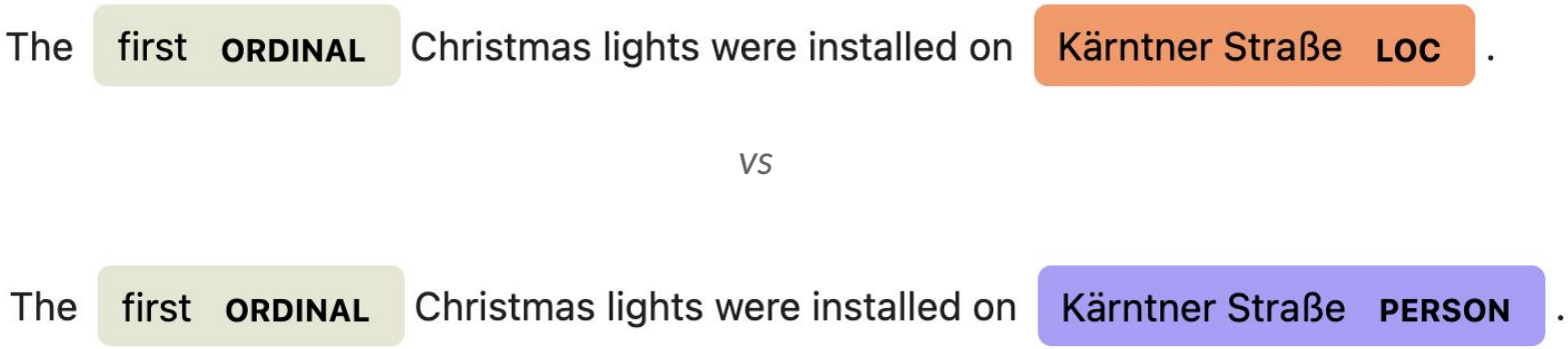
Choice of Language Data and Pipeline



Example: How spaCy's small English pipeline [above] works for place names in German

And what about the other way around? See the [below] example using spaCy's small German pipeline.

Model Size



Example: How spaCy's small English pipeline [above] and large English pipeline [below] work differently for the same sentence

Summary

- **Precision, Recall, and F1** score are commonly used metrics for evaluating geoparsing performance.
- Open questions in geoparsing evaluation include **the effects of input context, language choice, and model size**.

Lab Session

Goals

- ❏ We will learn how to compute geoparsing evaluation metrics.
- ❏ We will evaluate the micro-level behavior of spaCy models for geoparsing.
- ❏ We will learn how to **construct a ground-truth dataset for evaluation**.

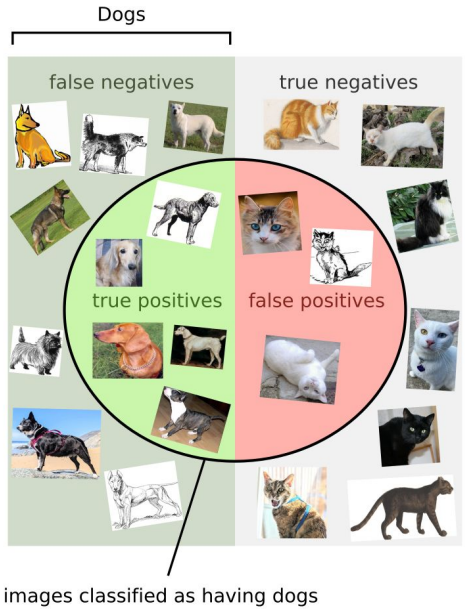
Submission

- Feel free to raise your hands to ask questions.
- Work independently but feel free to discuss with your colleagues and us.
- Make sure to upload your **.ipynb file** to Moodle by 23:59 today for your participation grade.
- In your Jupyter notebook, **include your code and answers with a clear format**.

Announcement

Recommended Reading

- A Wikipedia article about **Precision and recall**:
  https://en.wikipedia.org/wiki/Precision_and_recall



$$\text{Precision} = \frac{\text{5 true pos.}}{\text{8 total pos.}} \qquad \text{Recall} = \frac{\text{5 true pos.}}{\text{12 total dogs}}$$

The Upcoming Quiz

- We will have a quiz in the next class.
- It will take place after the (short) lecture and lab sessions.
- Please bring a pen.
- **Duration**: 45 minutes for 30 questions
- **Content coverage**: Theories and maths from Lectures 2 and 3; programming from Labs 2 and 3
- **Question format**: Multiple-choice questions

Week	Group Tue	Group Fri	Topic
Week 1	Oct 6	Oct 9	Introduction
Week 2	Oct 13	Oct 16	Geoparsing I
Week 3	Oct 20	Oct 23	Geoparsing II
Week 4	Oct 27	Nov 13	Geocoding I + **Quiz (Save the date!)**

Sample Questions

- Theory

1. Which of the following best describes the relationship between Geoparsing and Named Entity Recognition (NER)?
A. Geoparsing is a superset of NER.
B. Geoparsing is a subtask of NER focused on location references.
C. NER is a subtask of Geoparsing.
D. A and C

- Programming

7. In spaCy, which command is used to select the small English trained pipeline?
A. `spacy.load("en_core_web_sm")`
B. `spacy.import("en_core_web_sm")`
C. `spacy.use("en_core_web_sm")`
D. `spacy.init("en_core_web_sm")`

- Math

12. A geoparsing model processed a text and identified 15 place names. Upon manual checking, 10 of these were correct, and 5 were incorrect (e.g., person names mislabeled as GPE). There were 3 true location entities in the text that the model completely missed. What is the Precision?

A. 10/18

B. 10/15

C. 13/15

D. 10/13